Artificial Neural Network (ANN) for Transient-Based Leak Detection in Water Distribution Networks

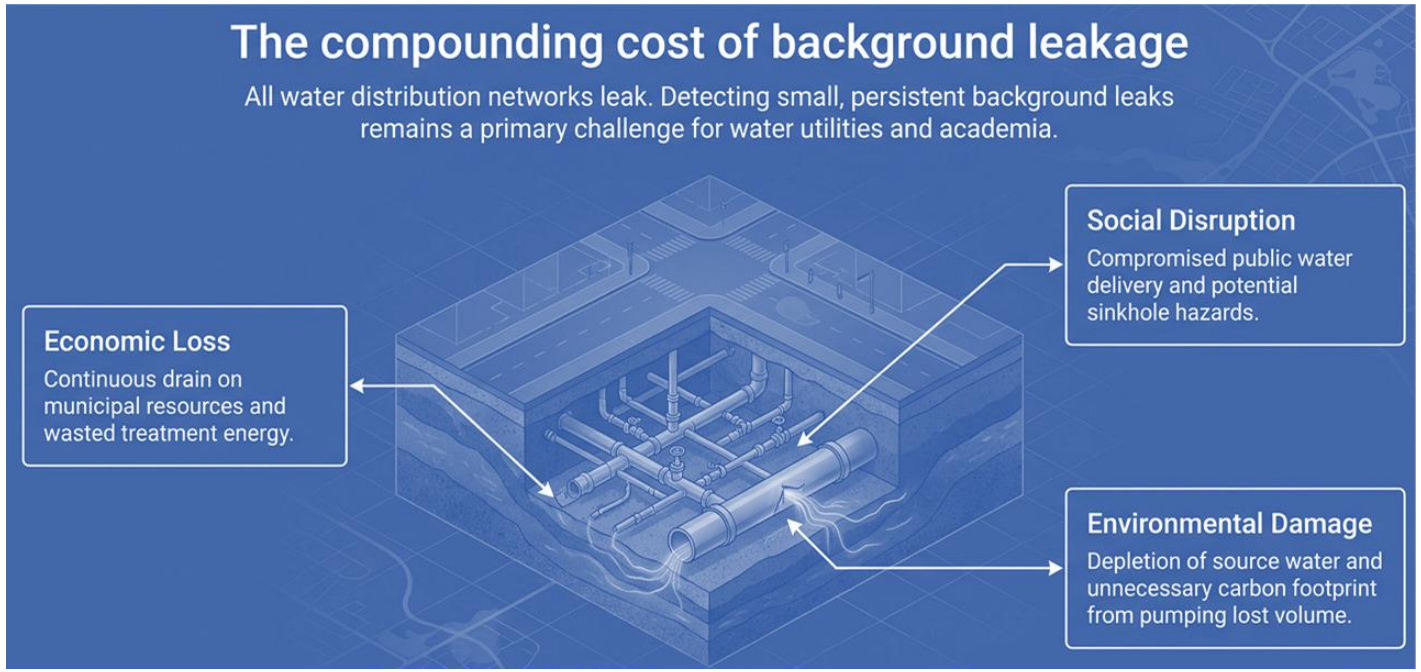
Ali Haghighi

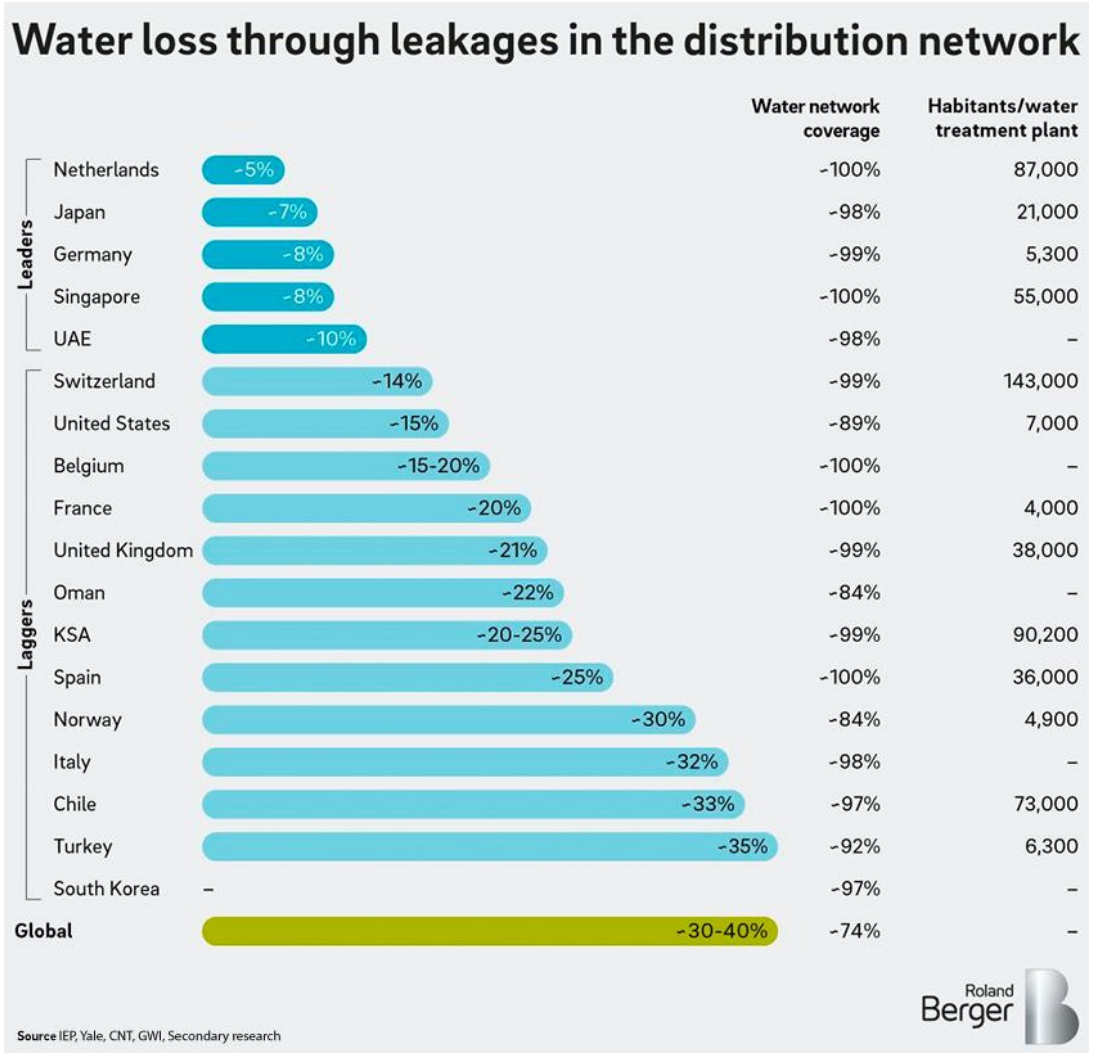
Department of Civil Engineering,
Technical University of Kaiserslautern (RPTU)
IAHR Committee on Transient Flows

Hosted by Spain Water and IWHR, China

03.11.2026, Kaiserslautern

Leak and its costs in water distribution networks



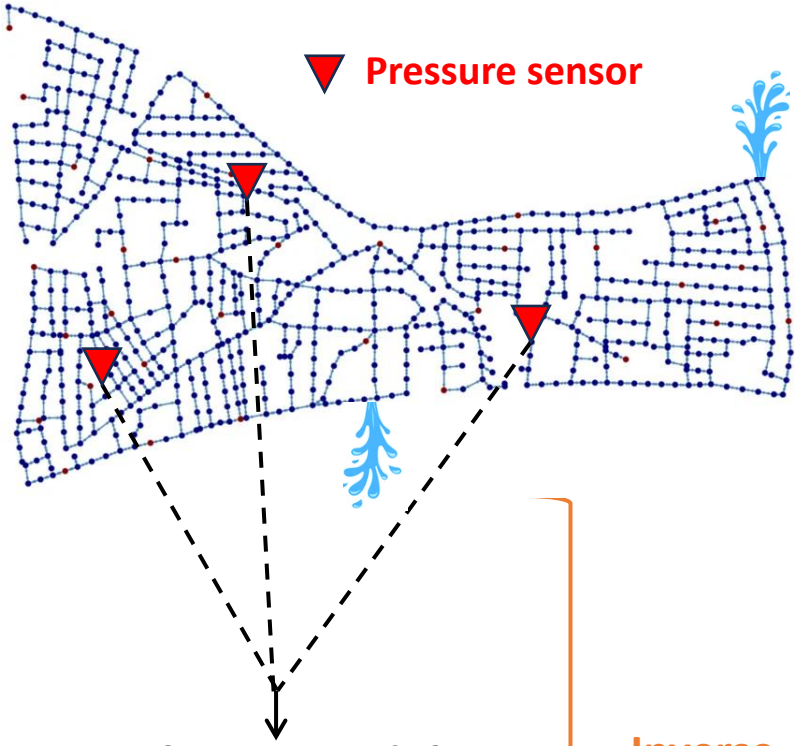


Leak detection approaches:

- Acoustic methods
- Systematic field inspections
- Satellite imagery
- Hydraulic-model-based techniques

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Hydraulic-model-based techniques



Simulation model

Optimization model

Data-driven (AI) model

↓

Unknowns like leaks

Inverse problem

INVERSE: measurements → unknown leaks

Forward problem

→ Simulation model →

Hydraulic responses:

H at heads

Q and V in pipes

Reversing the flow of computation

	The Forward Problem (Network Design & Analysis)	The Inverse Problem (Leak Detection)
Knowns	Physical characteristics (pipe lengths, diameters, friction factors, topology), water demands, boundary conditions.	Limited sensor data (a few nodal pressures and pipe flow rates).
Unknowns	The hydraulic state of the network.	The causes of the response (number, location, and magnitude of leaks, closed valves).
Direction of Computation	Parameters → Governing Equations → Predicted State	Hidden Parameters ← Inference ← Partial State

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Solving the Inverse Problem: Steady-State vs. Transient Flow

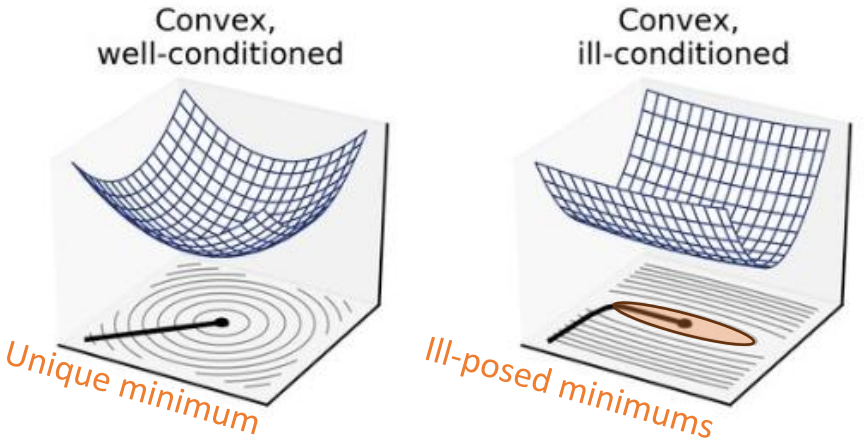
NLP of leak detection:

Model-predicted

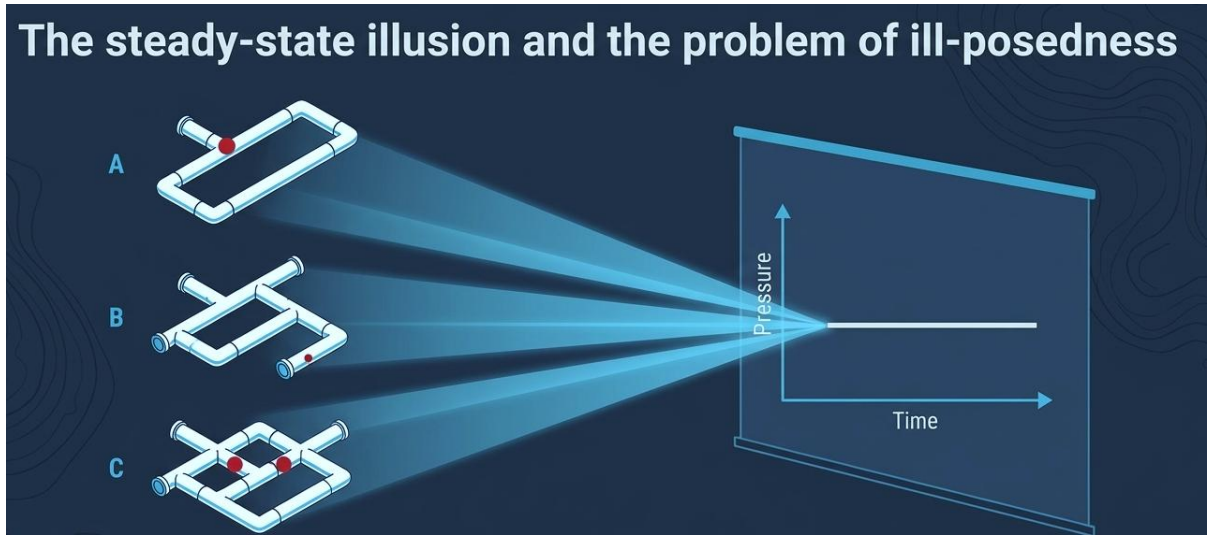
Observation

Minimize: $OF = \sum_{i=1}^{n_{sensor}} (H_i^m - H_i^o)^2$

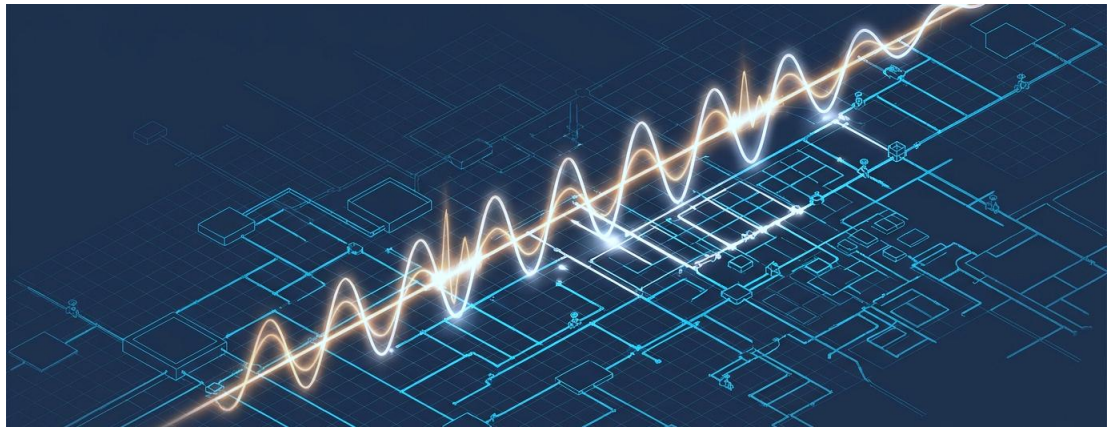
A nonlinear heuristic optimization like GA



Decision variables: Leak parameters (the number, location and size of leaks



Steady-State



Transient Flow

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Inverse Transient Analysis (ITA) for System Identification and Anomaly Detection

Transient simulation; Governing Equations

The Coupled PDEs (Quasi-Linear Hyperbolic):

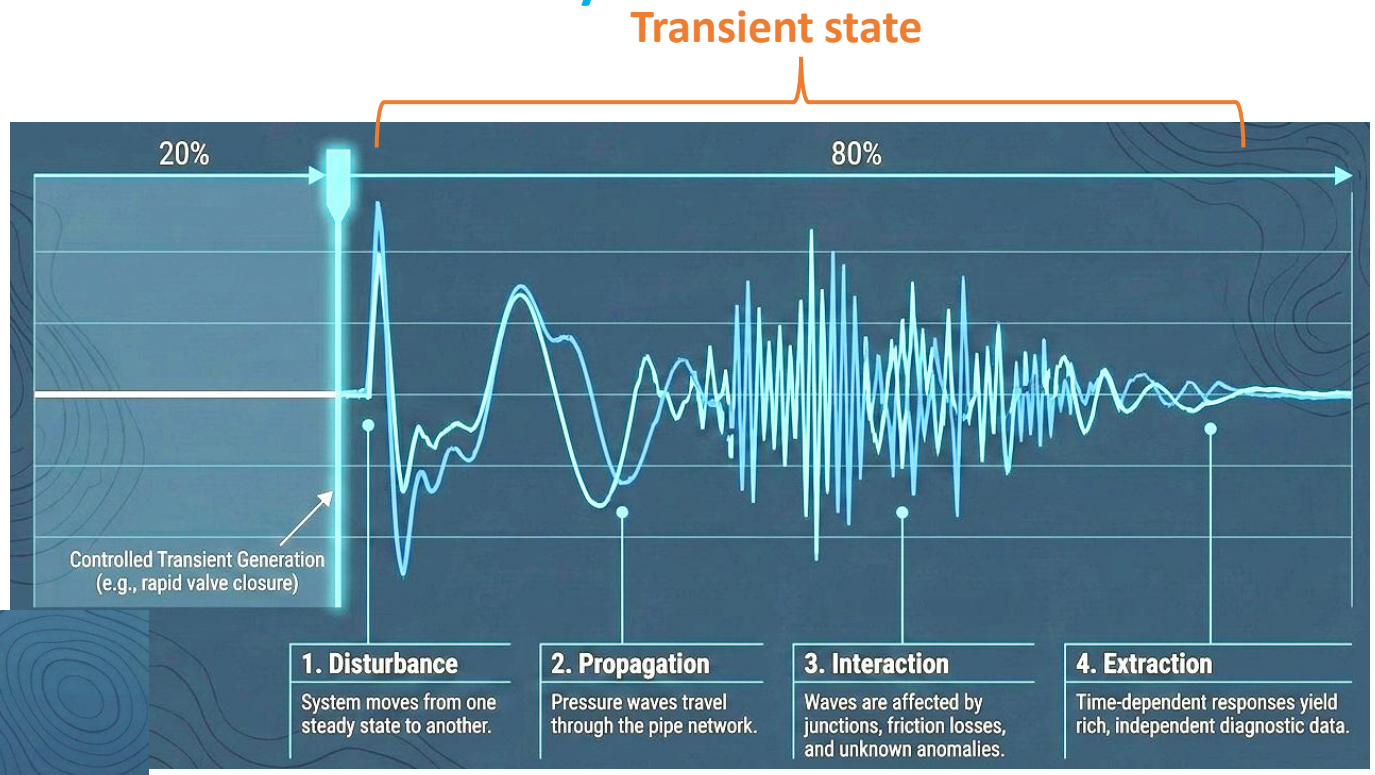
- Conservation of Momentum:

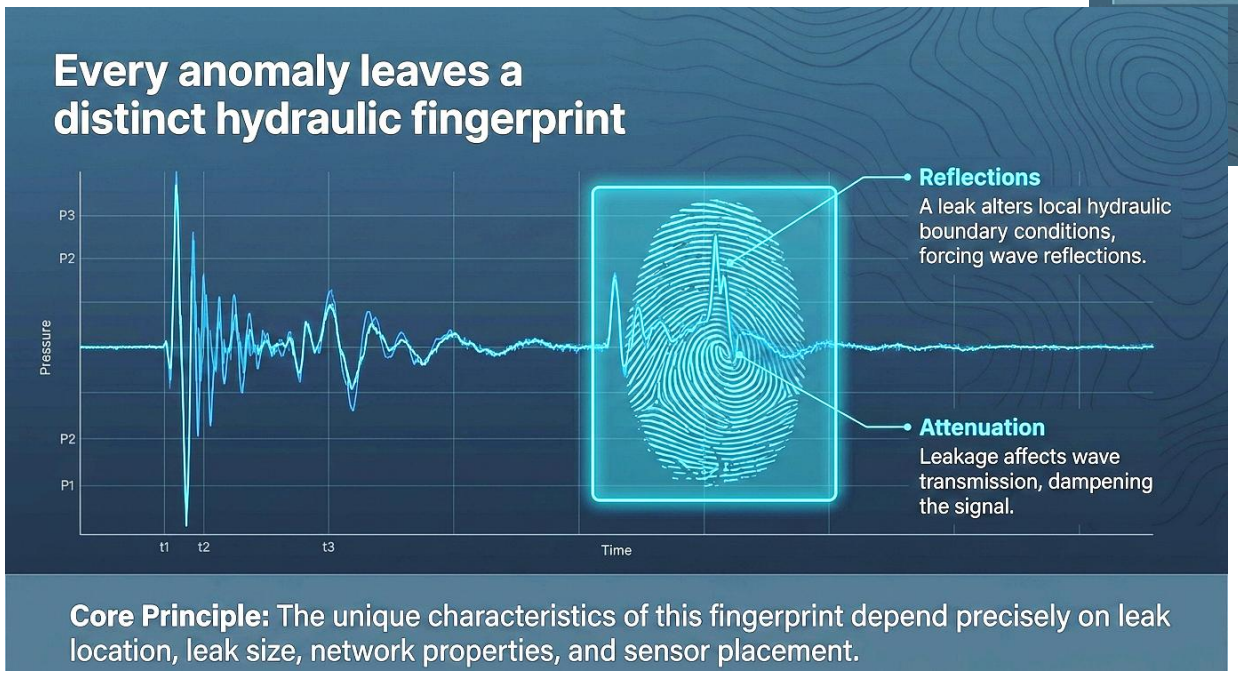
$$\frac{\partial H}{\partial x} + \frac{1}{gA}\frac{\partial Q}{\partial t} + \frac{fQ|Q|}{2gDA^2} = 0$$

⇐ **The MoC**

- Conservation of Mass:

$$\frac{\partial Q}{\partial x} + \frac{gA}{a^2}\frac{\partial H}{\partial t} = 0$$





This is the fundamental idea behind the Inverse Transient Analysis (ITA):

By exploiting these transient signatures, we can obtain additional information about the network's condition and potentially reduce the non-uniqueness and identifiability problems associated with steady-state leak detection.

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The ITA implementation for a water distribution network

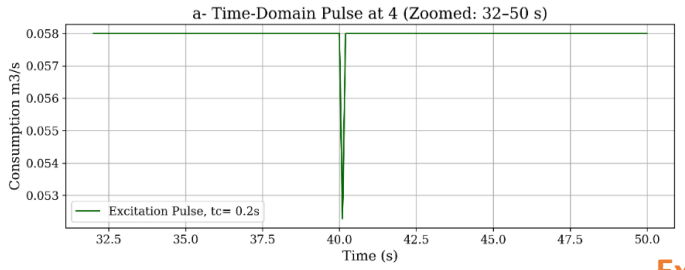
Step 1. Measurement Site Design

Step 2. Development of a Transient Hydraulic Model

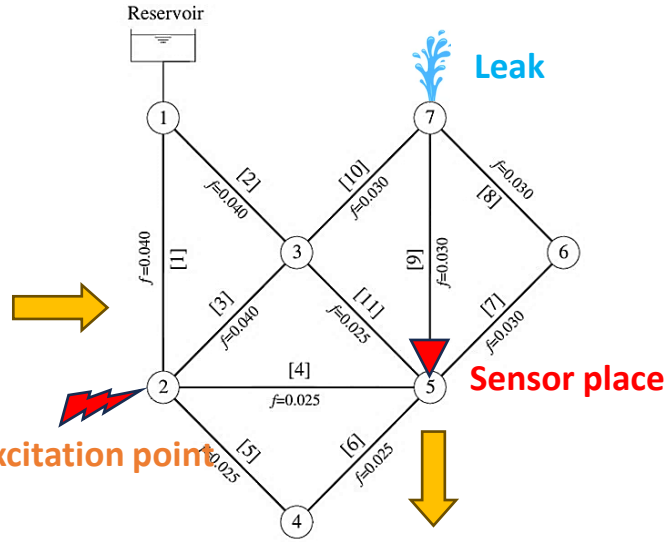
Step 3. Controlled Transient Generation and Measurement

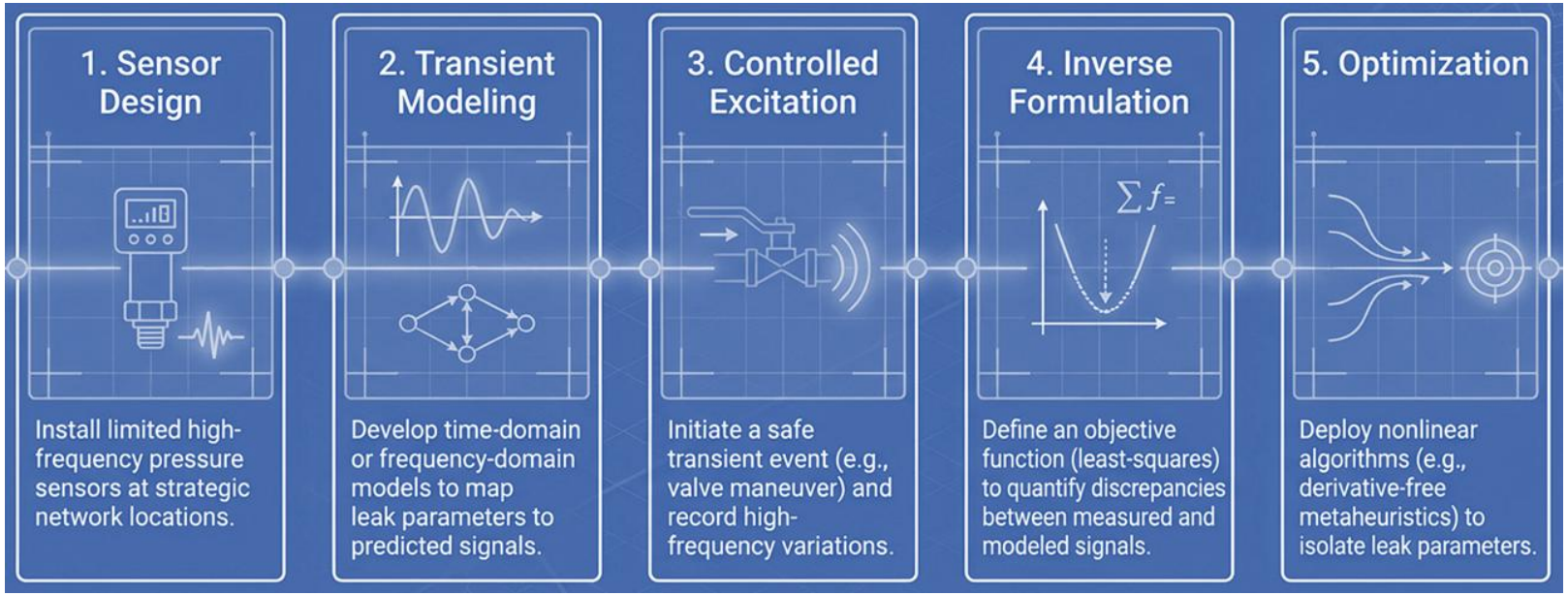
Step 4. Formulation of the Inverse Problem

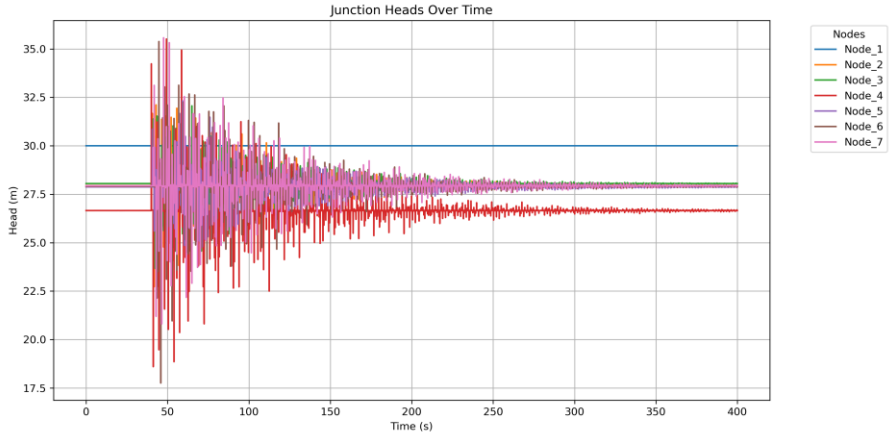
Step 5. Optimization and Leak Identification



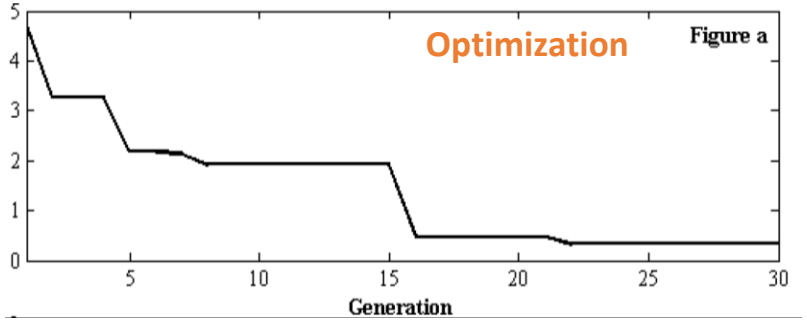
Pulse







Optimization



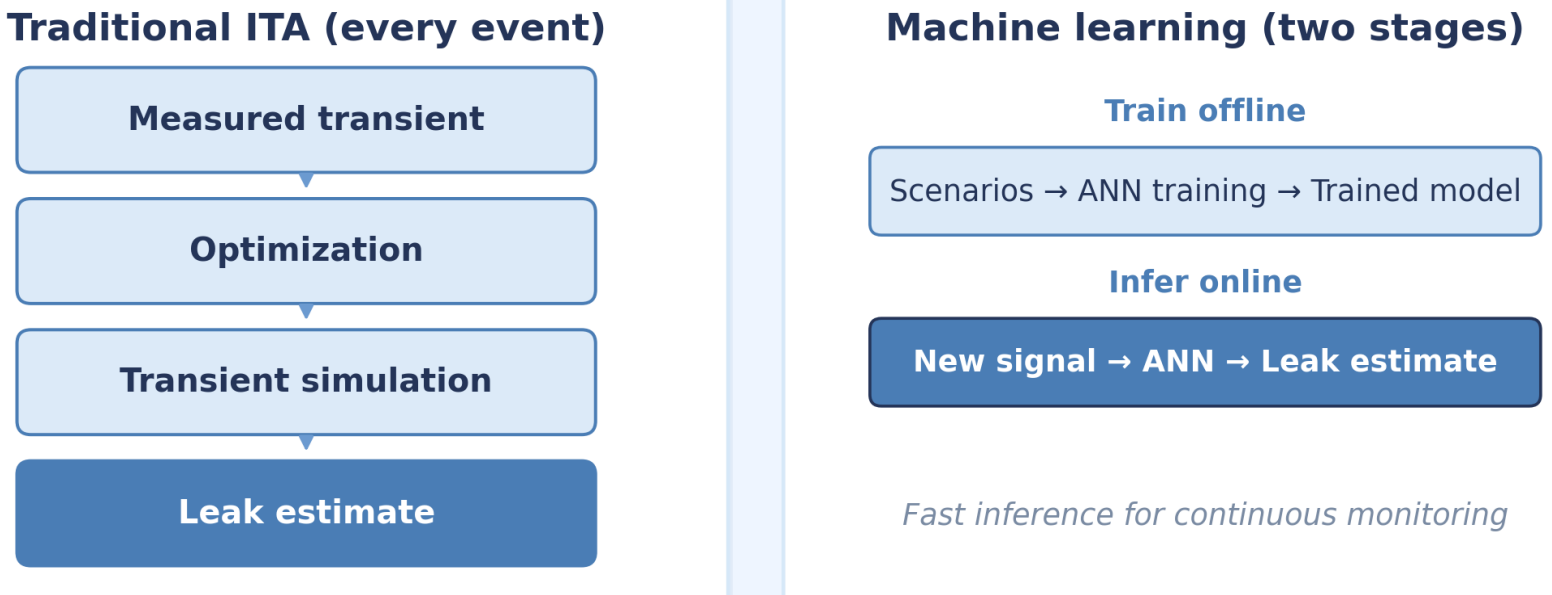
Machine Learning as an Alternative for Solving ITA Problems

Challenges of Traditional ITA:

1. **Dependence on an Accurate and Deterministic Models**
2. **Measurement Noise and Uncertainty**
3. **High Real-Time Computational Cost**
4. **Changing Network Conditions**
5. **Dependence on the Transient Excitation**
6. **Highly Nonlinear Relationship Between Leaks and Transient**
7. **High-Dimensional Transient Signals**

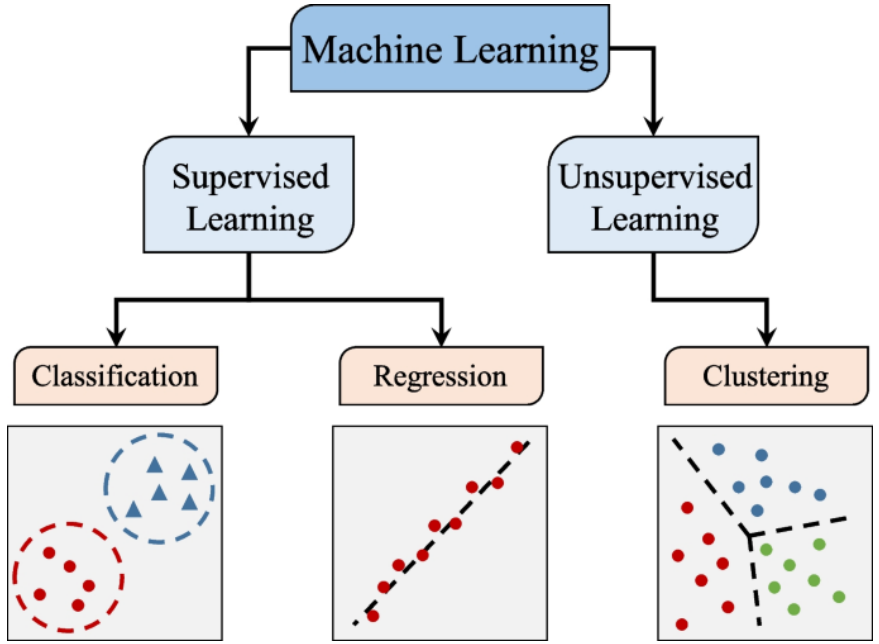
From ITA to Machine Learning

The solutions are in stochastic data-derive, probabilistic and machine learning approaches!



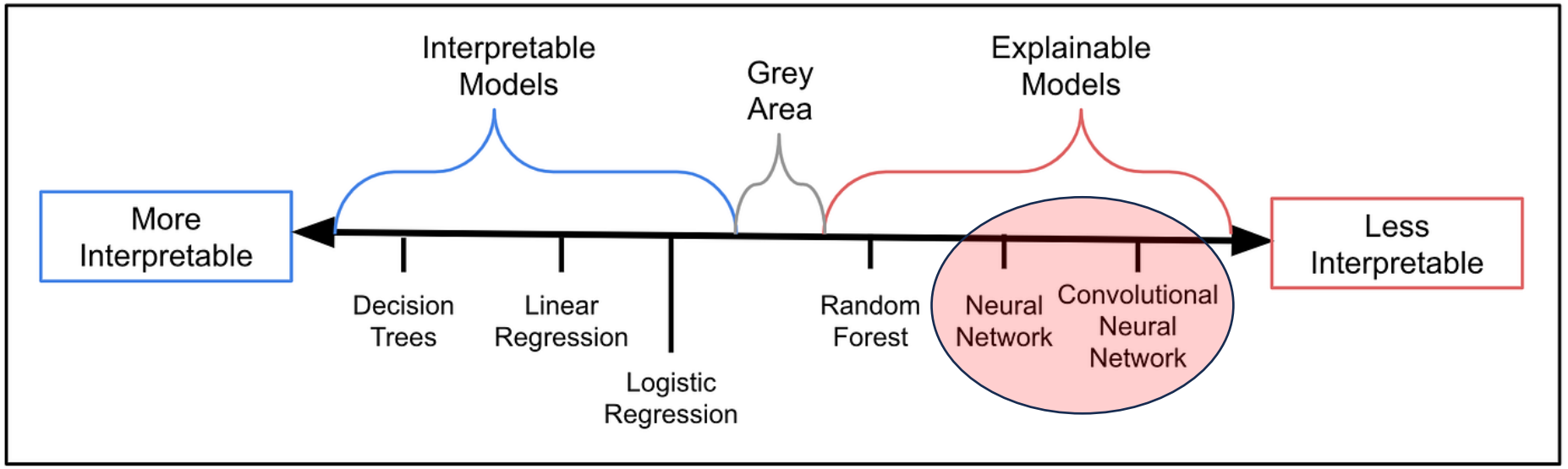
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An Introduction to Machine Learning to Leak Detection



In our case, a machine-learning model could learn to:

- Classify whether a leak is present.
- Identify the leak location.
- Estimate the leak magnitude.
- Distinguish different hydraulic or operational conditions.



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What is an Artificial Neural Network?

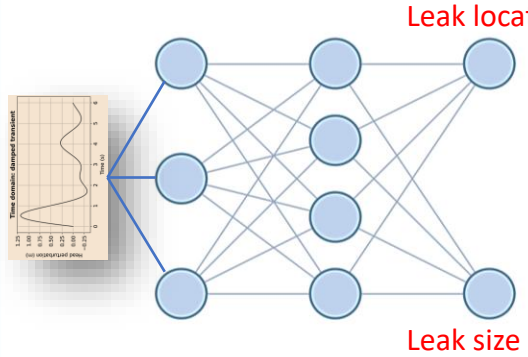
A flexible model that learns x → y from labelled examples

1 Key ideas

- Inputs are features from transient pressure and flow.
- Outputs are leak-related labels or continuous estimates.
- The network learns patterns present in the training data.
- Hydraulic understanding still guides features and validation.

Concept diagram

x → ANN → y



The artificial neuron

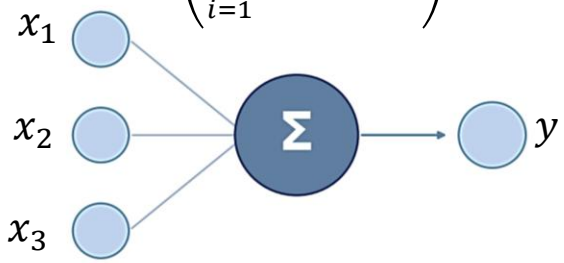
Weighted sum, bias, and nonlinear activation

2 Key ideas

- Each connection carries a learnable weight.
- A bias term shifts the decision surface.
- Nonlinearity enables complex leak–signal maps.
- Many neurons stacked in layers form an MLP.

Concept diagram

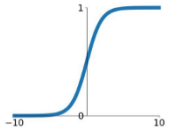
$$y = f\left(\sum_{i=1}^{n}(w_i x_i) + b\right)$$



Activation Functions

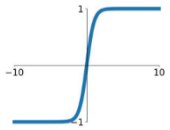
Sigmoid

$\sigma(x) = \frac{1}{1+e^{-x}}$



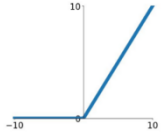
tanh

$\tanh(x)$



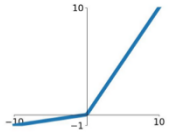
ReLU

$\max(0, x)$



Leaky ReLU

$\max(0.1x, x)$

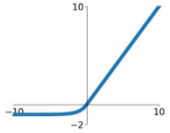


Maxout

$\max(w_1^T x + b_1, w_2^T x + b_2)$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Commonly used activation functions *f*

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A Multilayer Perceptron (MLP), fully connected ANN

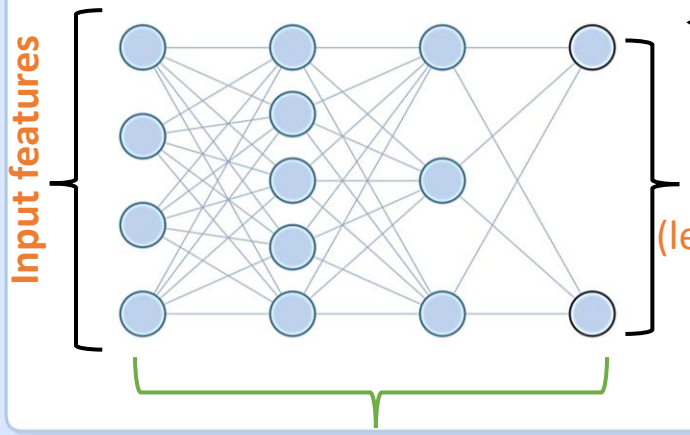
Multilayer Perceptron

A practical starting architecture for hydraulic features

3 Key ideas

- Input layer sized to the feature dimension.
- One or two hidden layers are often enough.
- Output layer matches detection, location, or size.
- Prefer better features over unnecessary depth.

Concept diagram



Back propagation to adjust *w* and *b*

Labels → **Loss function**

(leak parameters)

It learns the nonlinear relationship between the input hydraulic features and leaks!

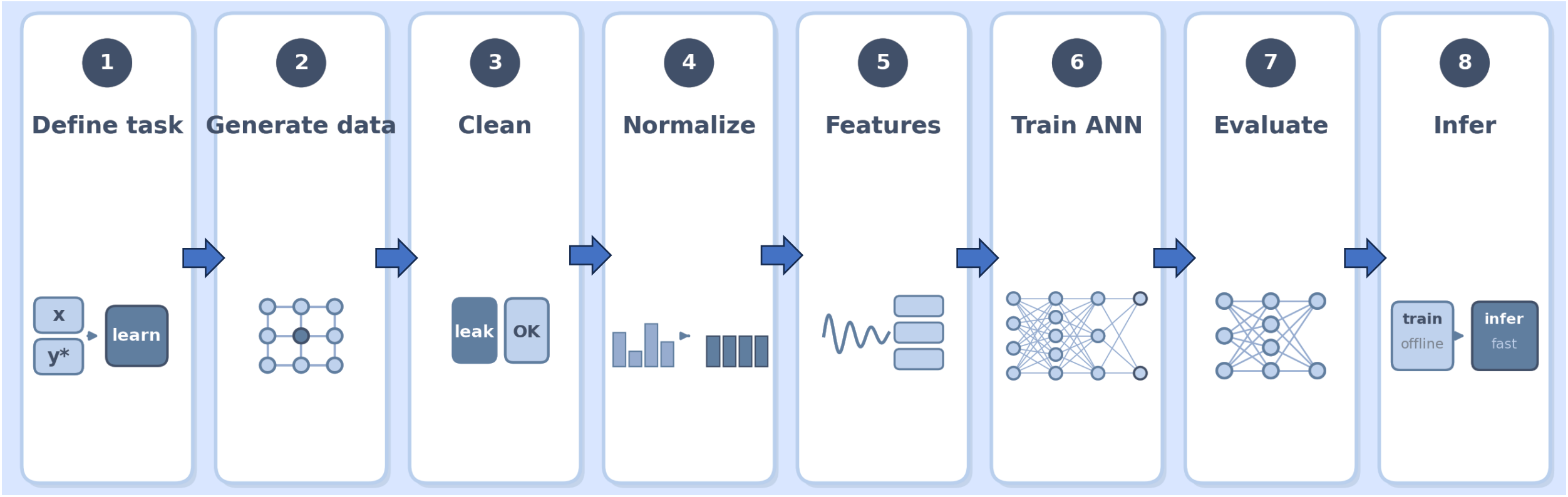
The leak detection goals with ML:

Detection: is a leak present? (classification)

Localization: where is the leak? (multiclass classification or zone identification)

Sizing: how large is the leak? (regression)

The Supervised Learning Idea for Leak Detection: End-to-End ANN Pipeline



- Labels
- Hydraulics
- Site design
- Simulation
- Transient test

- Leak scenarios
- Noise
- Uncertainties
- Dataset
- Real data

- Failed simulations
- Missing data
- Outliers
- Denoising
- Sensors alignment

- Preprocessing
- Normalization
- Standardization
- Data leakage

- Feature extraction
- Dimensionality reduction
- Transformations

- ANN architecture
- Dataset split
- Loss function
- Backpropagation
- Overfitting

- Accuracy
- Error
- Precision/recall
- Confusion matrices
- Sensitivity to noise
- Deploying and Retraining

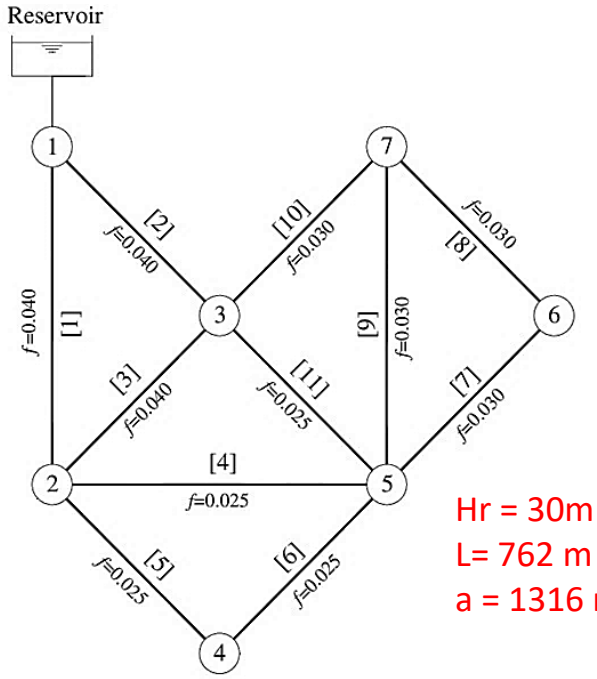
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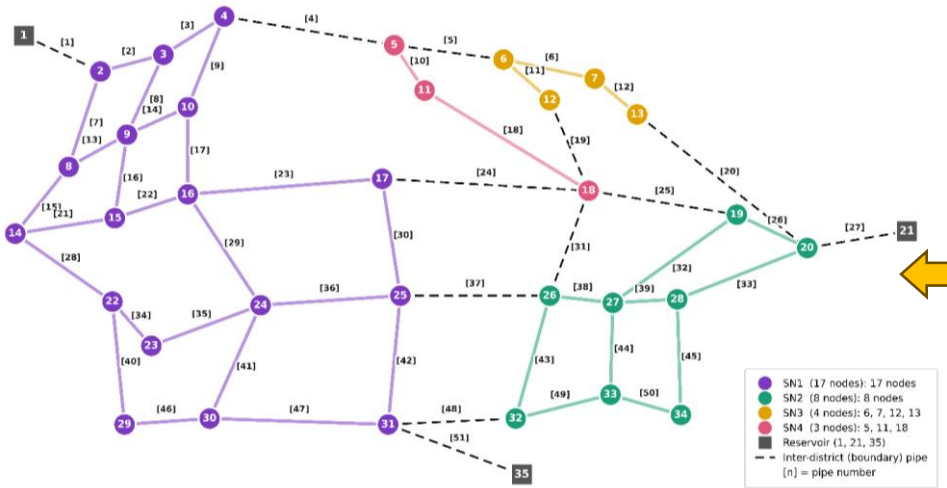
Thank you!

Water Distribution Network Tutorial



Hr = 30m
L= 762 m
a = 1316 m/s

Liggett, J. A., & Chen, L. C. (1994). Inverse transient analysis in pipe networks. Journal of Hydraulic Engineering, 120(8), 934–955.



Prerequisite Tasks for Transient-Based Leak Detection

1. Transient Simulation Model

- Method of Characteristics (MoC), time domain
- Transfer Matrix Method (TMM), frequency domain
- FFT and IFFT for conversion between time and frequency domains

2. Optimal Transient Excitation

- Identify the most suitable excitation location
- Select the excitation type and operating conditions
- Maximize the information content of the generated transient

3. Optimal Sensor Placement

- Identify the most informative measurement locations
- Evaluate leak detectability and sensor sensitivity
- Optimize sensor number and distribution

4. Network Decomposition

- Analyze transient propagation, attenuation, and damping
- Identify poorly observable or weakly excited network sections
- Decompose large networks into smaller, manageable sub-networks

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Designing the ITA problem

1. Transient Simulation Model
2. Optimal Transient Excitation
3. Optimal Sensor Placement

Water Management
Volume 164 Issue WM6

Transient generation in pipe networks for leak detection
Haghighi and Shamloo

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Transient generation in pipe networks for leak detection

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J. Hydraul. Struct., 2019; 5(2):32-48
DOI: 10.22055/jhs.2019.29402.1107

OPEN ACCESS

Transient Measurement Site Design in Pipe Networks Using the Decision Table Method (DTM)

Mohammad Hadi Ranginkaman[1]
Amir Houshang Ayati[2]
Amin Ebrahim Bakhshipour[3]
Ali Haghighi[4]

Abstract

The accuracy of leak detection and calibration of pipe networks by means of the inverse transient analysis (ITA) is highly affected by the number and location of the measurement sites. This study introduces a conceptual decision-making model, the Decision Table Method (DTM), for measurement site design of pipe networks with the aim of inverse transient analysis. Through the Decision Table Method, near optimum measurement sites are decided based on two criteria of the maximum sensitivity of measurement sites and the maximum diversity of sensitivity with

Check for updates

ASCE

Multiobjective Wrapper Sampling Design for Leak Detection of Pipe Networks Based on Machine Learning and Transient Methods

Amir Houshang Ayati[1] and Ali Haghighi[2]

Abstract: This study introduces a novel sampling design (SD) method for hybrid Machine Learning/Transient-Based (ML/TB) leak detection of pipe networks. The proposed technique exploits the hydraulic responses of the network in the frequency domain and the concept of Filter and Wrapper feature selection as a machine learning approach. It also utilizes multiobjective optimization to handle the trade-off between leak detection error and number of sampling nodes. To apply this method, a transient hydraulic simulation model of the network is developed in the time domain. Then, considering a wide range of leak scenarios, the hydraulic responses of the network are calculated at candidate measurement sites. The responses are then transferred into the frequency domain using the Fast Fourier Transform (FFT) and stored as the train and test datasets. To reduce the dimensions of the initial feature vector, a threshold is applied to the responses to filter the very high frequencies. Finally, a classifier based on a Linear Discriminant Algorithm (LDA) coupled to a binary-coded Non-dominated Sorting Genetic Algorithm (NSGA-II) is applied to preprocessed datasets. Four sampling design methods are adopted from the literature and modified in the frequency domain for more investigations and comparisons. Solving two example pipe networks showed that the proposed method outperforms the existing approaches with respect to higher accuracy leak detection with fewer sampling sites. **DOI: 10.1061/JWRMD5.WRENG-5620.**
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Author keywords: Leak detection; Machine learning (ML); Sampling design (SD); Feature selection; Multiobjective optimization.

Optimal Measurement Site Locations for Inverse Transient Analysis in Pipe Networks

John P. Vítkovský[1]; James A. Liggett[2]; Angus R. Simpson, M.ASCE[3]; and Martin F. Lambert[4]

Abstract: The quality of leak detection and quantification and calibration for friction coefficients in pipelines and networks by the inverse transient method are dependent on the quantity and location of data measurement sites. This paper presents an approach for determining the configuration of measurement sites that produces optimal results. Three performance indicators, two that are based on A- and D-optimality criteria and one that is based on the sensitivities of the heads with respect to the parameters, show which configurations are superior. These are illustrated by two case studies, the first of which is a small pipe network in which all configurations are considered directly (fully enumerable) and the second is a larger pipe network in which statistics are drawn from a sampling of configurations. For the large network, a genetic algorithm, with a new crossover operator, performs a search of possible measurement site configurations to determine the optimal measurement locations. The number of sites as well as time length of data at each site are also considered.

DOI: 10.1061/(ASCE)0733-9496(2003)129:6(480)

CE Database subject headings: Pipe networks; Leakage; Friction coefficients; Locations; Data collection; Methodology.

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Hydraulic Impedance Spectroscopy

1- A transient simulation model of the WDN is created using the MOC or TMM

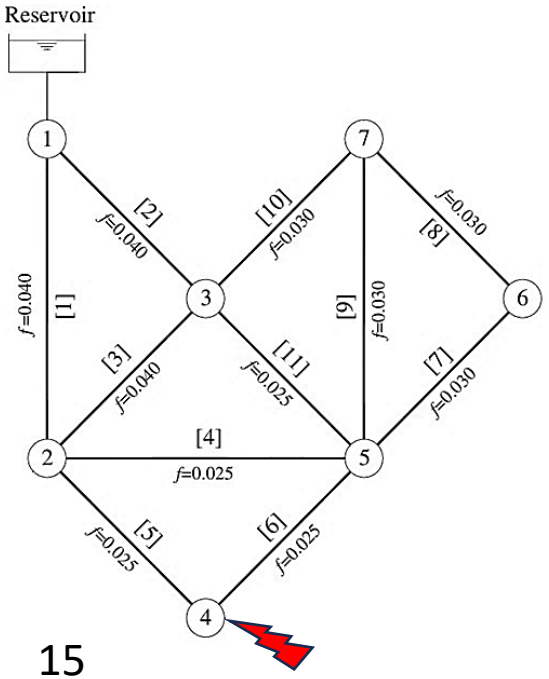
2- A fast triangular impulse is applied as the transient excitation at each node of the WDN to induce a transient state.

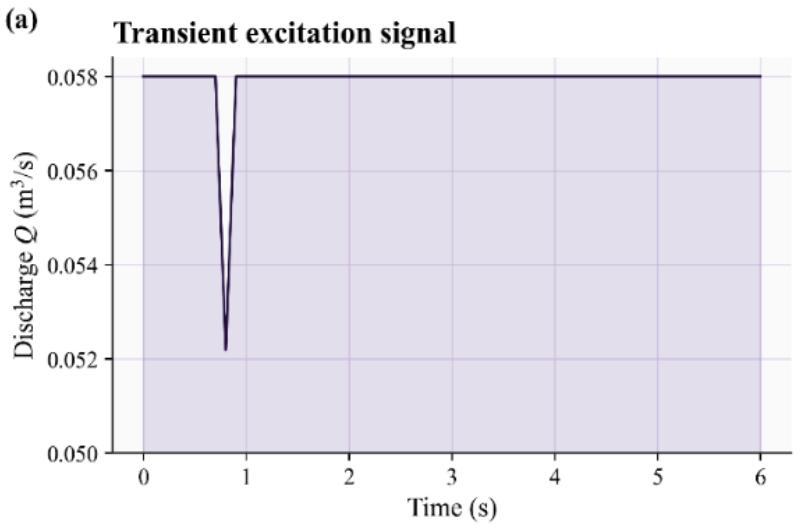
3- Take the FFT of the input impulse: $q(t) \rightarrow q(\omega)$

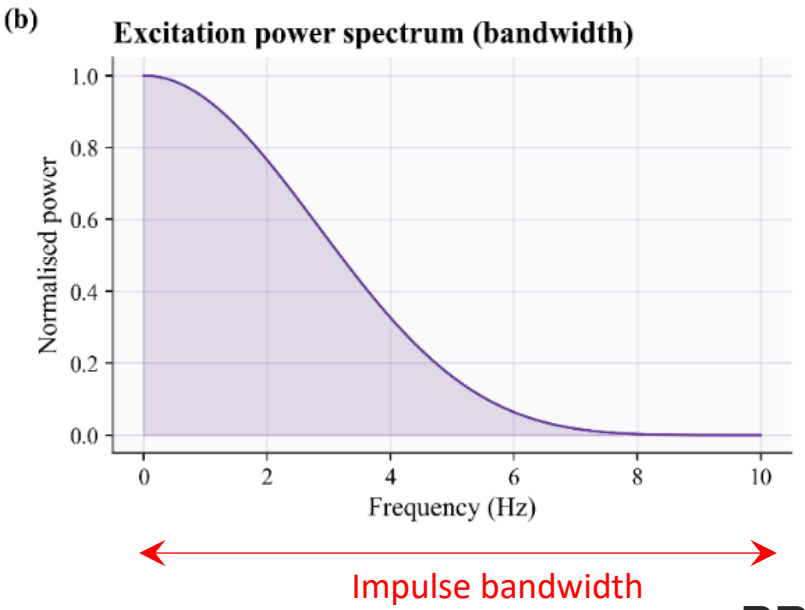
4- The nodal frequency responses $h(\omega)$ of the system are analyzed using the simulation model for every impulse location scenario.

5- The Frequency Response Function (FRF) of each node $\boldsymbol{H(\omega) = \frac{h(\omega)}{q(\omega)}}$ is computed.

Then, the nodal power spectrum is obtained as $\boldsymbol{P(\omega) = |H(\omega)|^2}$







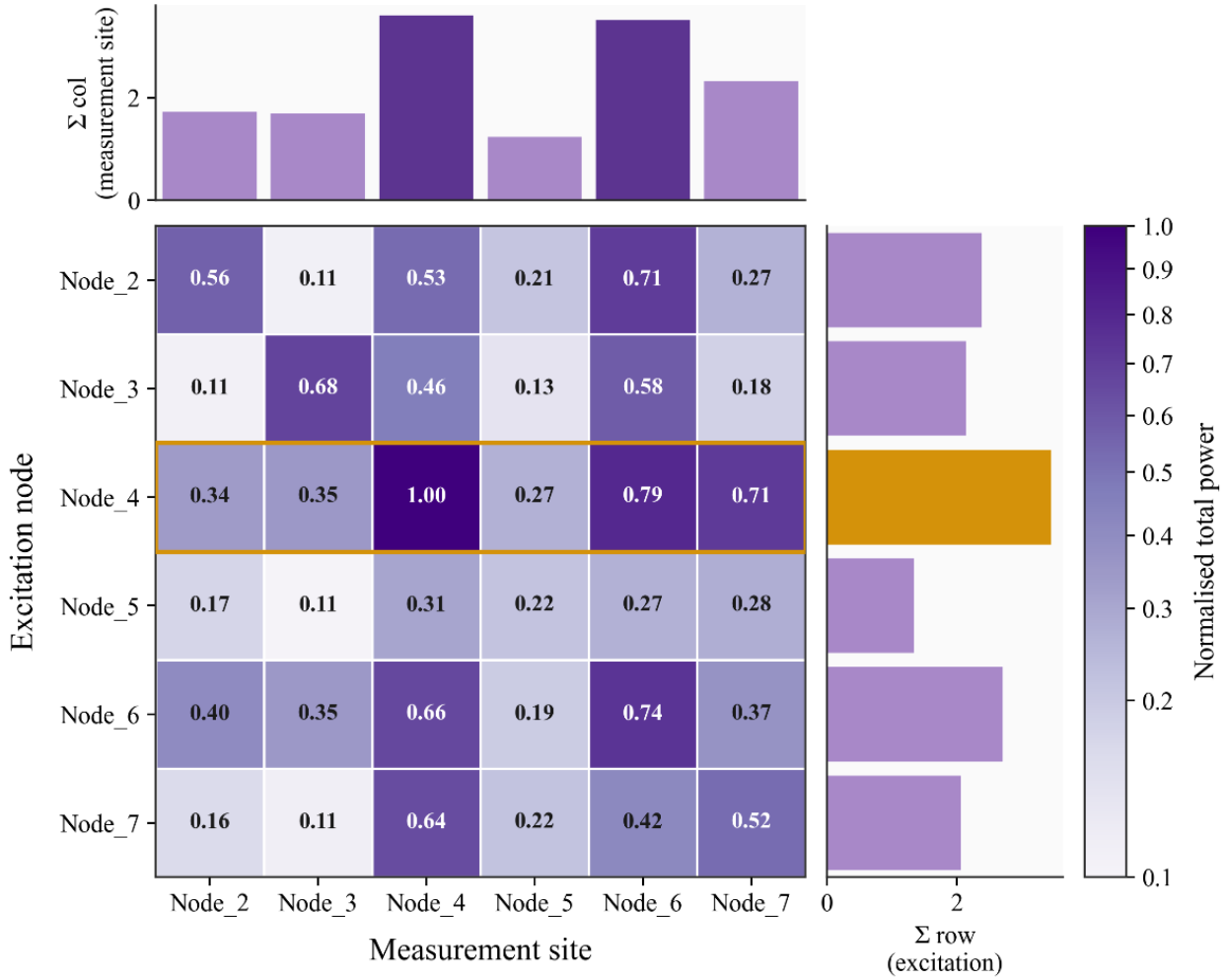
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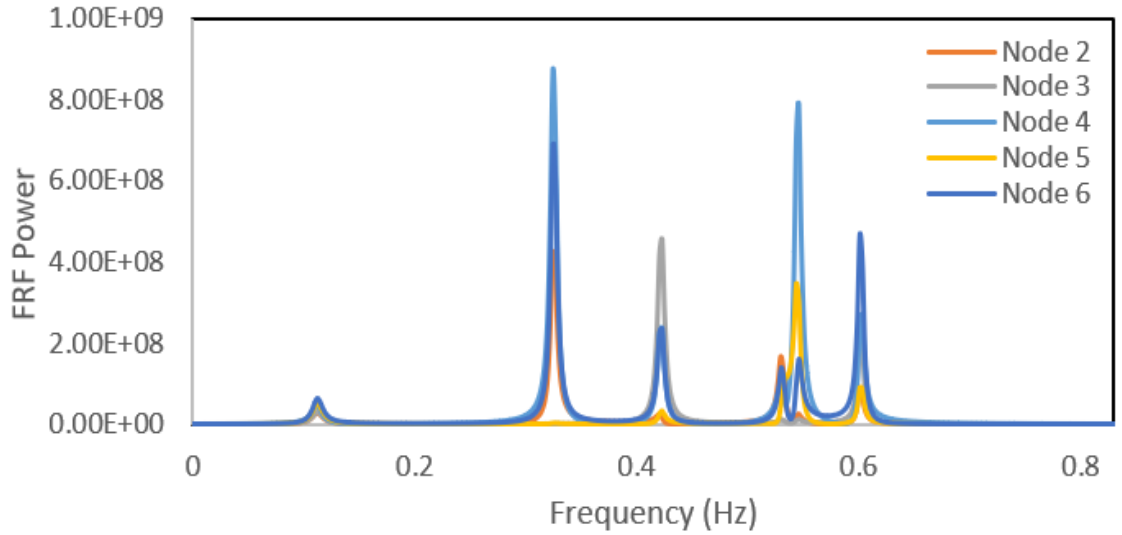
Hydraulic Impedance Spectroscopy

6- For every node we compute the FRF with respect to the impulse at every node resulting in an $n \times n$ **excitation-response matrix**.

7- Identifying key **Resonant frequencies:** Using a peak detection algorithm applied to the FRF spectrum for the interval (0-0.875 Hz), the resonant frequencies are identified as **[0.11, 0.32, 0.42, 0.53, 0.54, 0.60]** Hz.

8- **Power spectrum heatmap:** The total power of each node's power spectrum is computed, averaged and normalized

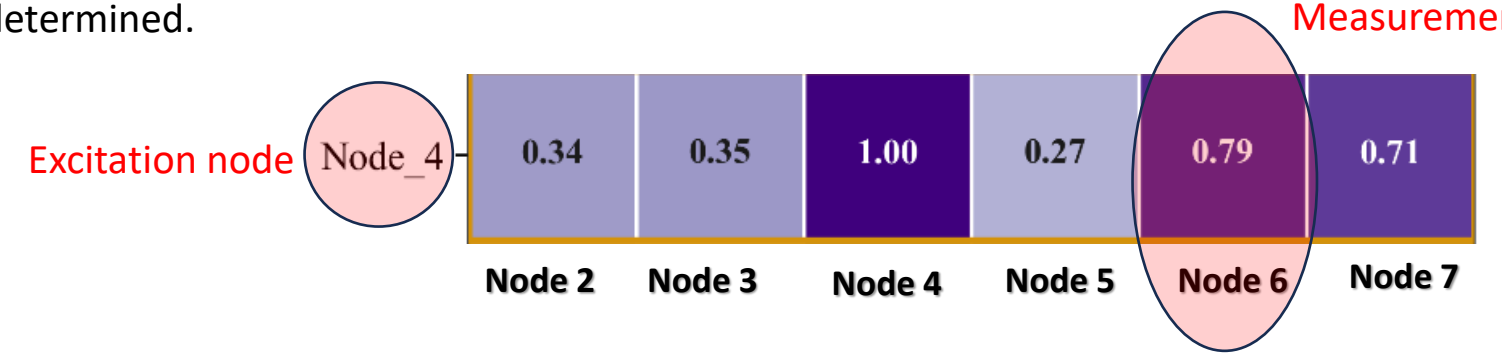




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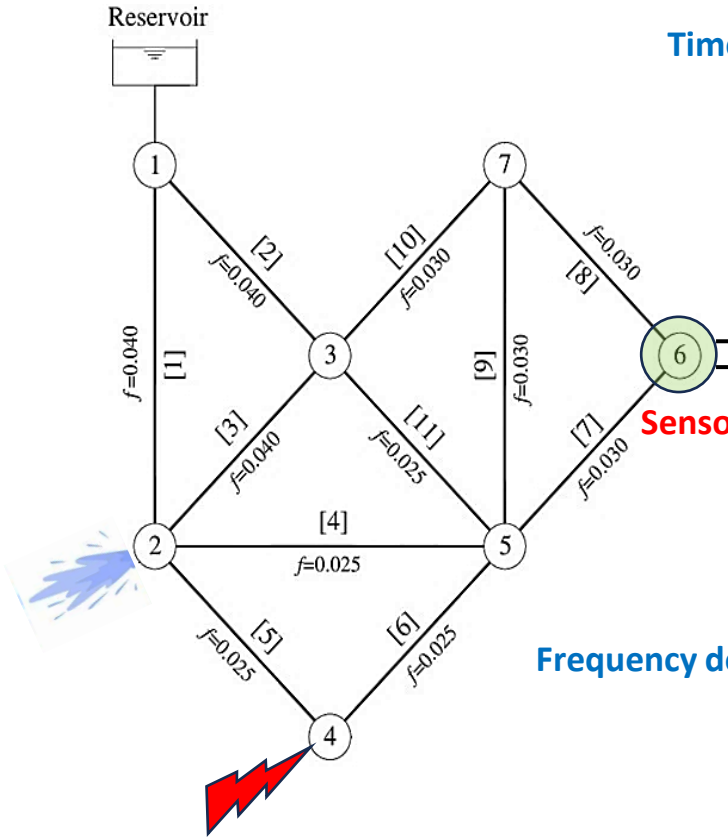
Hydraulic Impedance Spectroscopy

Supposing the transient is initiated from Node 4, focusing on the row of Node 4 in the heatmap, the optimum locations for sensor placements are explicitly determined.



Number of sampling sites	The best measurement site configuration using:				
	HIS approach	**DTM** (Ranginkaman et al. 2019)	**η_j indictor** (Vitkovsky et.al 2003)	**η_A indictor** (Vitkovsky et.al 2003)	**η_D indictor** (Vitkovsky et.al 2003)
1	4	6	6	4	4
2	4,6	4,6	4,6	4,6	4,6
3	4, 6, 7	4, 6, 7	4, 6, 7	3, 4, 6	4, 6, 7
4	2, 4, 6, 7	4, 5, 6, 7	4, 5, 6, 7	3, 4, 6, 7	3, 4, 6, 7
5	2, 4, 5, 6, 7	2, 4, 5, 6, 7	2, 4, 5, 6, 7	2, 3, 4, 6, 7	3, 4, 5, 6, 7
6	2, 3, 4, 5, 6, 7	2, 3, 4, 5, 6, 7	2, 3, 4, 5, 6, 7	2, 3, 4, 5, 6, 7	2, 3, 4, 5, 6, 7

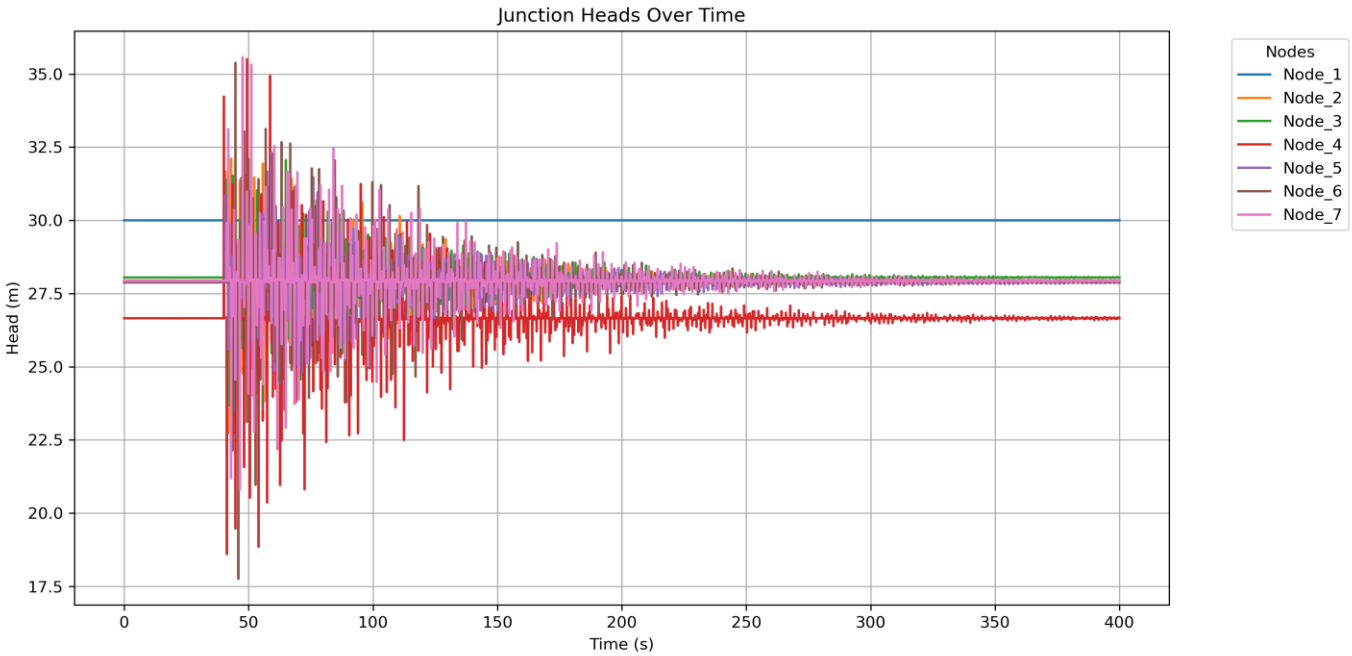
Feature engineering for ANN

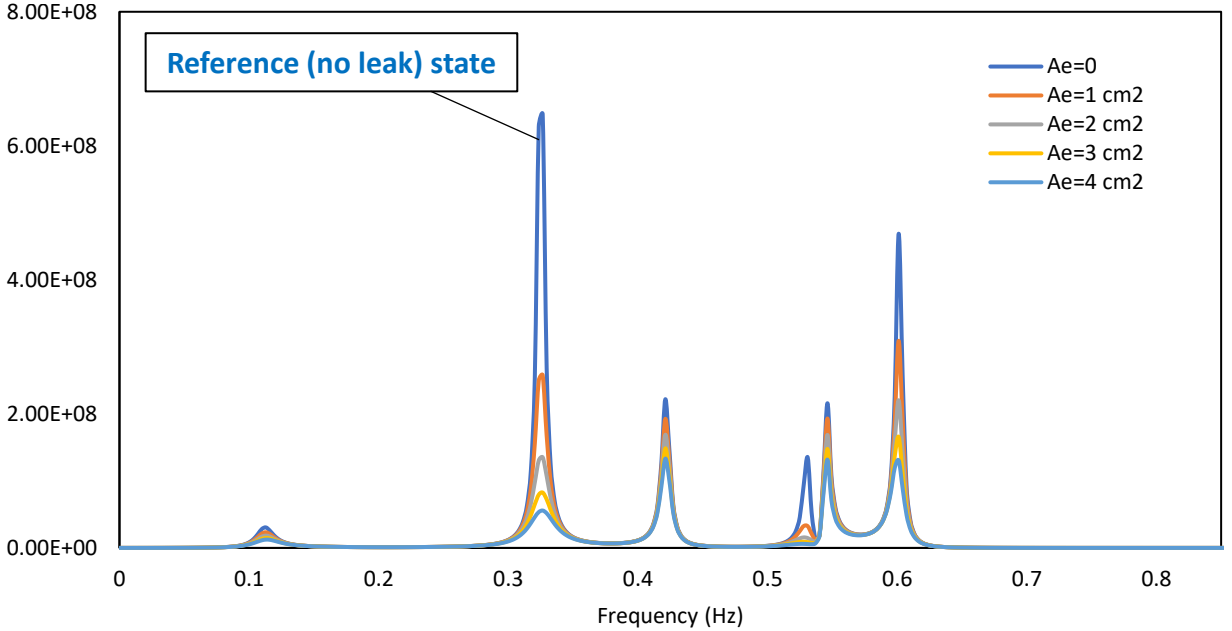


Time domain signal

Sensor

Frequency domain spectrum

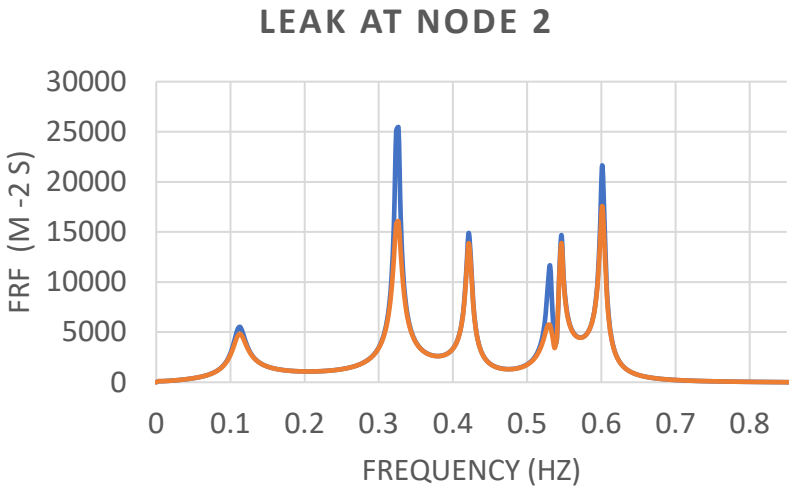


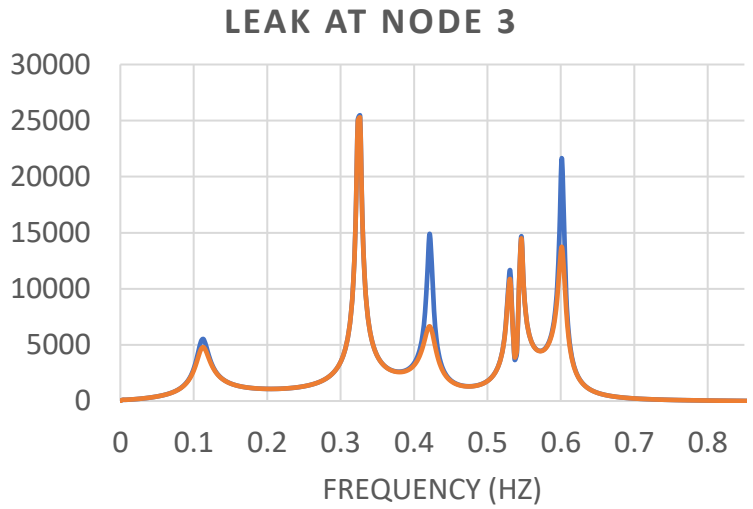


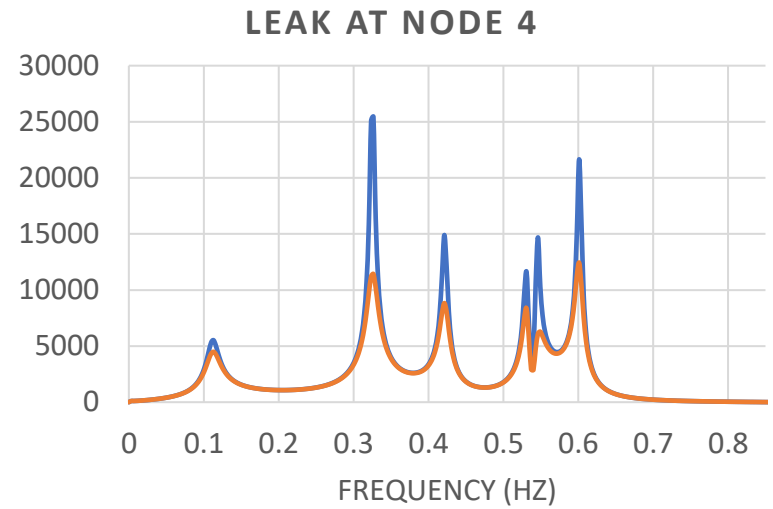
Features: Powers at resonant frequencies [0.11, 0.32, 0.42, 0.53, 0.54, 0.60]

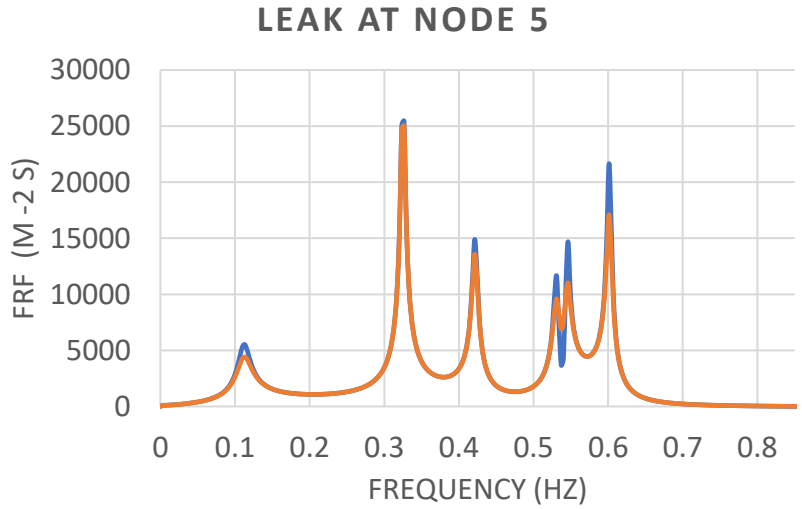
RPTU

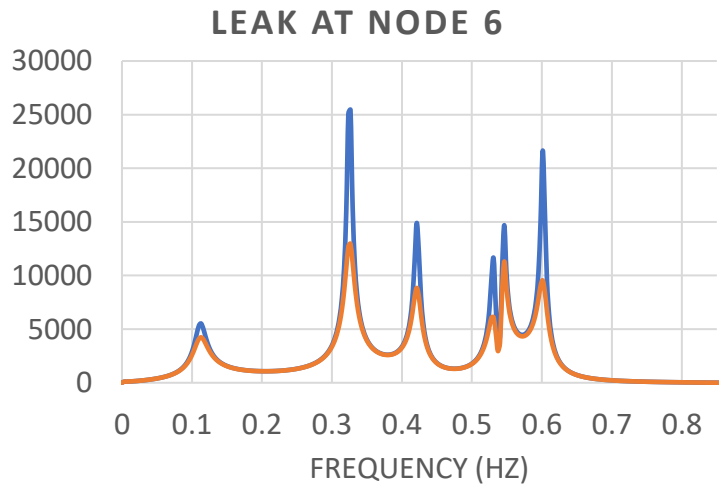
Leakage fingerprints on the FRF of the measurement site, Node 6

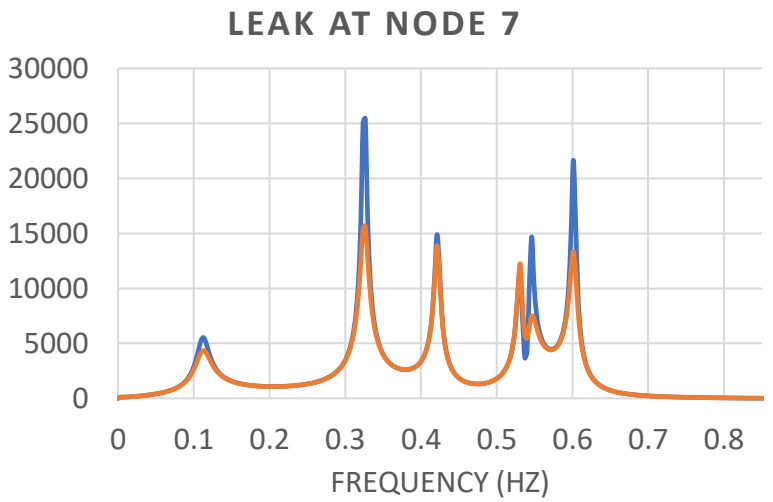












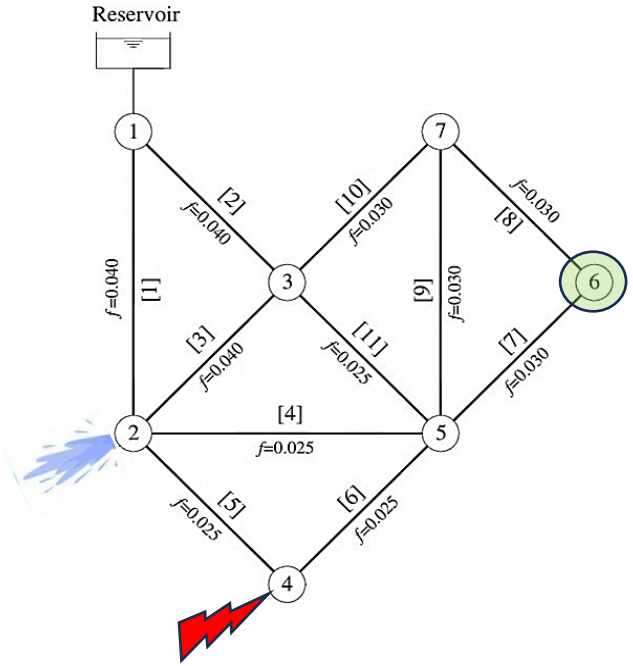
RPTU

Dataset generation

Leak parameters: The number, Location and **Size of Leaks**

Dataset Settings	
Excitation node: 4	Leak areas (cm²): 0.1-4
Sensor node(s): 6	Relative noise= 5% Augmentation replicates=10
Leak / output junctions: 2,3,4,5,6,7	Seed: 100

Final Structured Dataset	
Number of rows (samples): 310	ANN output: 6 leak parameters (existence or size)
Number of features: 6	Reference rows (no leak samples): 10
Types of features: Resonant frequencies	Leak rows (leak scenarios): 300

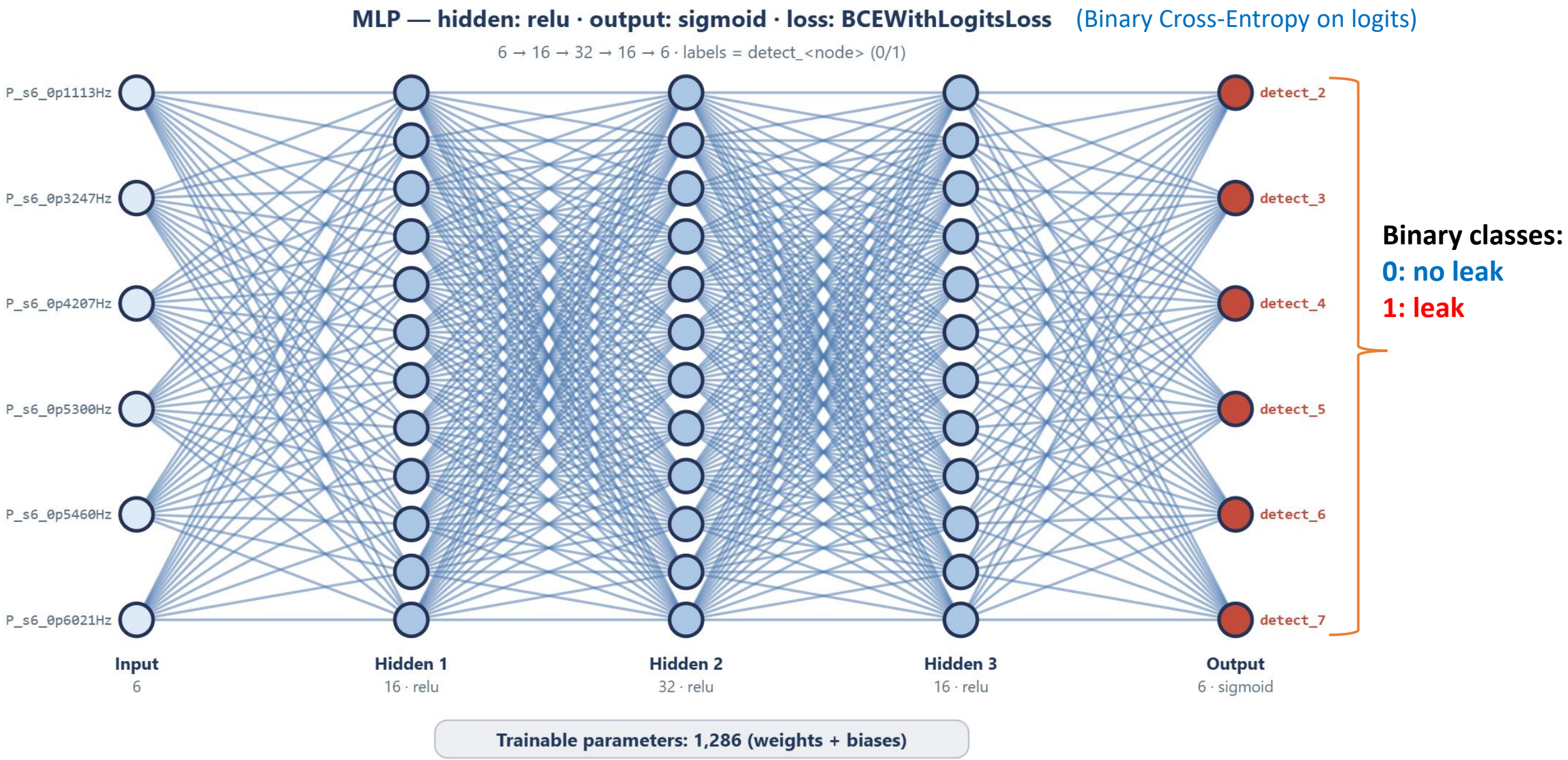


Labels | **Features**

Node	leak_present	leak_size_mm	P_s6_0.1113Hz	P_s6_0.3247Hz	P_s6_0.4207Hz	P_s6_0.5300Hz	P_s6_0.5460Hz	P_s6_0.6021Hz
none	0	0.00	3.01E+07	7.62E+08	2.14E+08	1.37E+08	2.21E+08	4.62E+08
2	1	3.57	2.97E+07	7.05E+08	2.12E+08	1.22E+08	2.20E+08	4.49E+08
2	1	7.98	2.80E+07	5.30E+08	2.07E+08	8.09E+07	2.15E+08	4.05E+08
2	1	11.28	2.61E+07	3.90E+08	2.01E+08	5.36E+07	2.09E+08	3.58E+08
2	1	15.96	2.29E+07	2.37E+08	1.89E+08	2.86E+07	1.94E+08	2.86E+08
2	1	22.57	1.80E+07	1.14E+08	1.69E+08	1.24E+07	1.67E+08	1.95E+08
3	1	3.57	2.97E+07	7.62E+08	1.87E+08	1.35E+08	2.21E+08	4.31E+08
3	1	7.98	2.79E+07	7.61E+08	1.17E+08	1.31E+08	2.20E+08	3.36E+08
3	1	11.28	2.60E+07	7.61E+08	7.32E+07	1.26E+08	2.18E+08	2.55E+08
3	1	15.96	2.26E+07	7.59E+08	3.64E+07	1.17E+08	2.15E+08	1.61E+08

RPTU

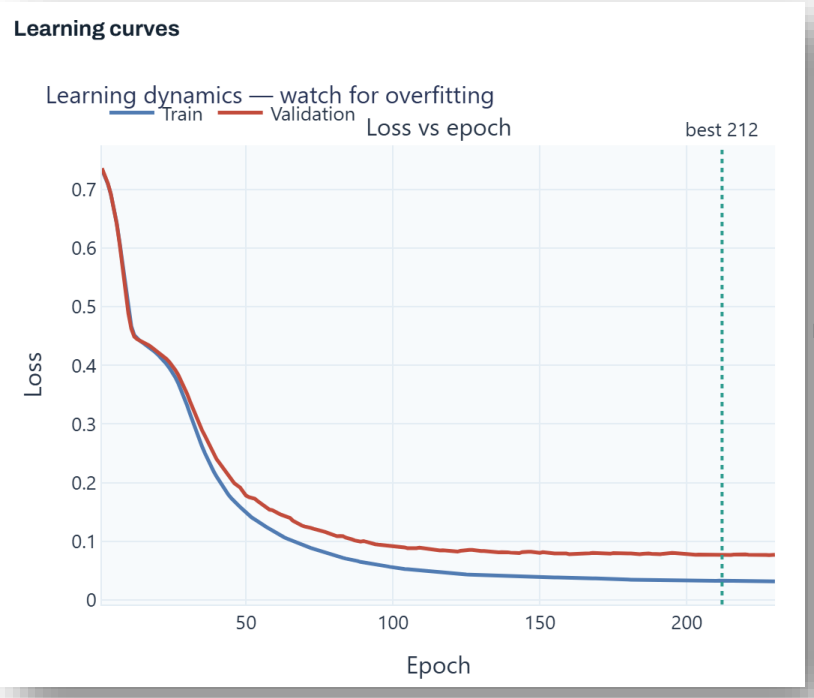
Model 1: Binary classification



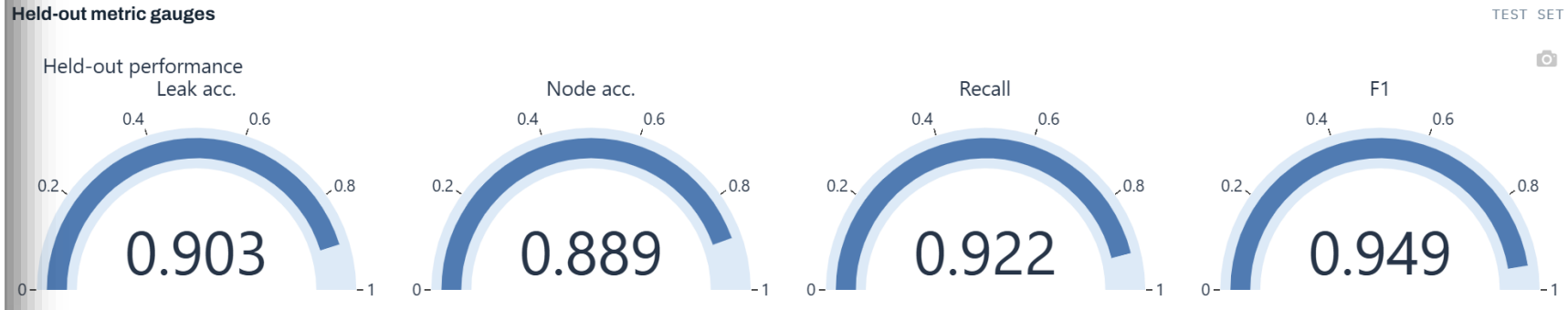
Binary classes:
0: no leak
1: leak

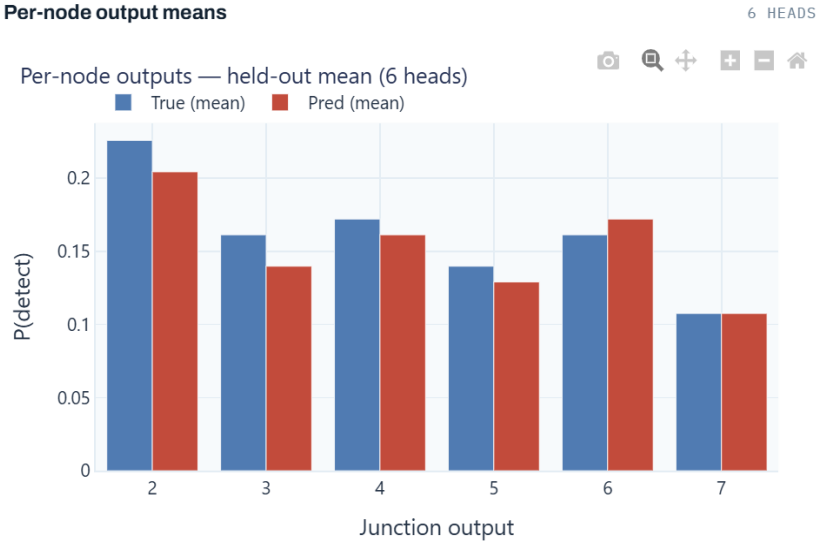
RPTU

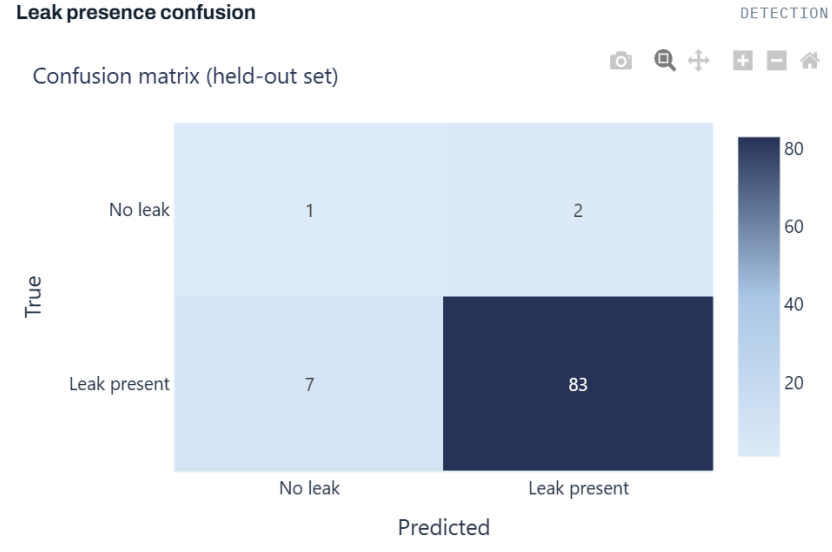
Model 1: Binary classification (yes/no classification)



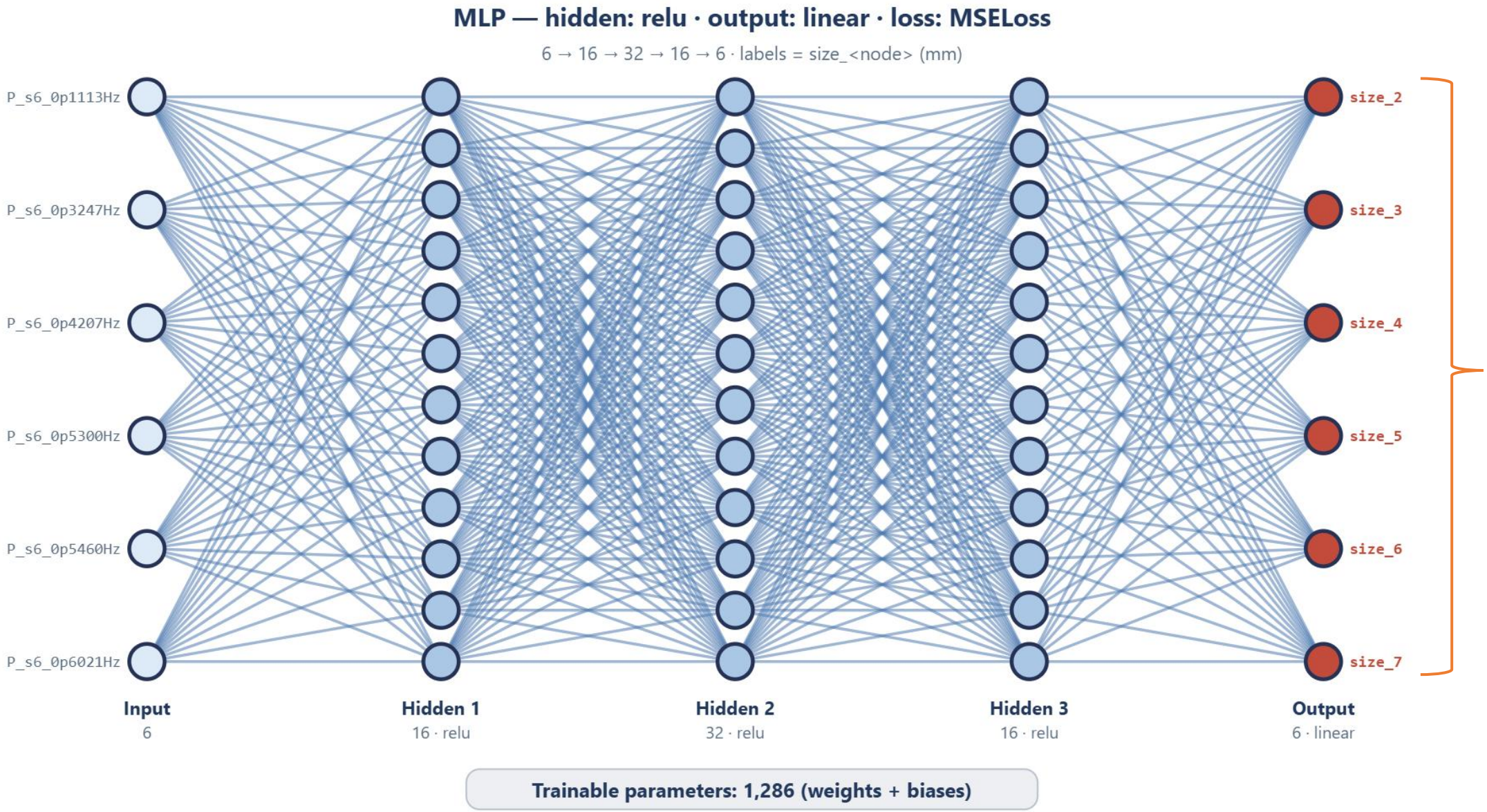
Learning Settings	
Learning rate: 0.001	Dropout: 15%
Test and Validation fraction: 30%	Batch size: 32
Early stopping= yes - patience 20	Framework: Torch







Model 2: Regression (leak sizing)

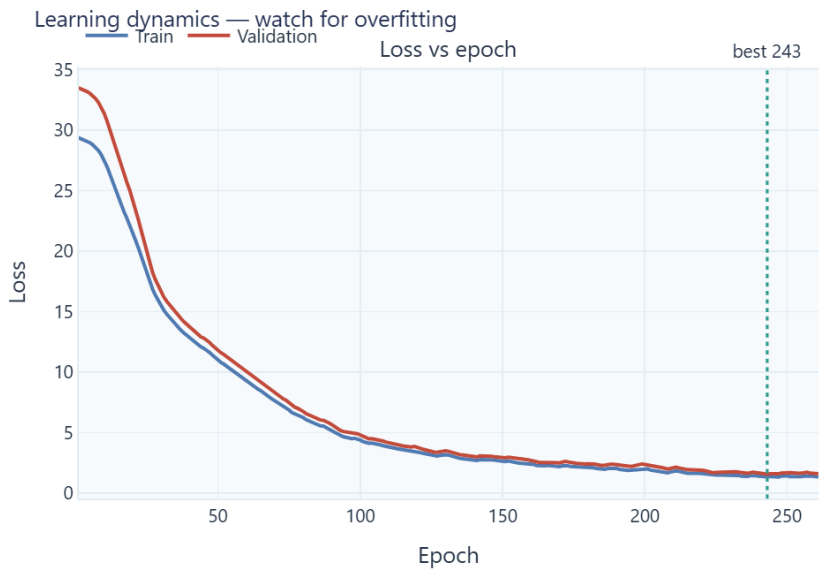


Real values:
leak sizes

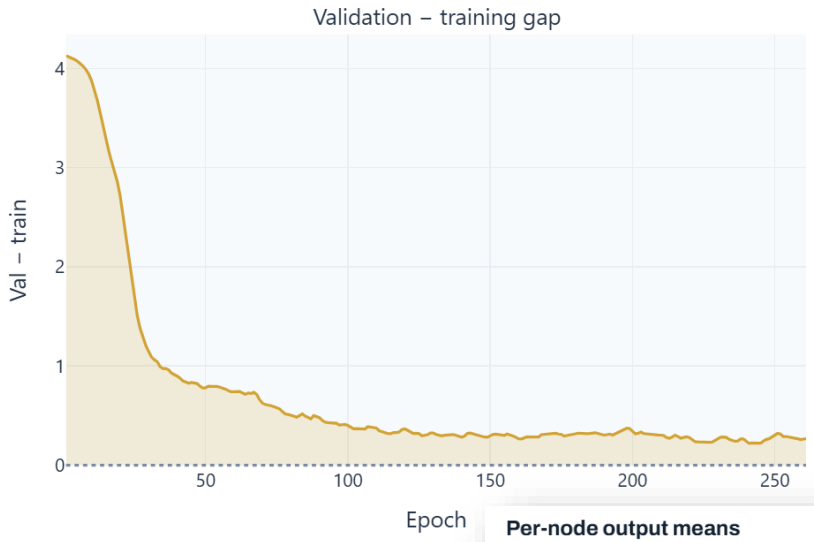
RPTU

Model 2: Regression (leak sizing)

Learning curves

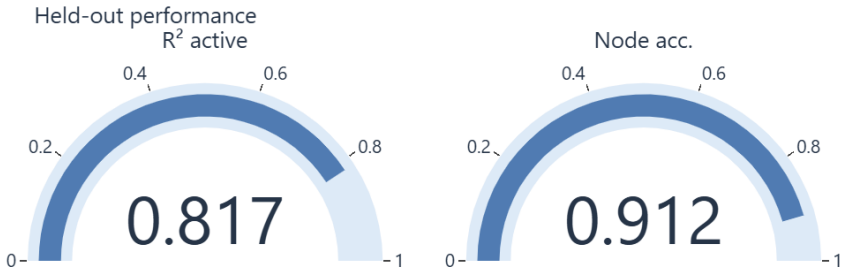


TRAIN VS VALIDATION



Learning Settings	
Learning rate: 0.001	Dropout: 15%
Test and Validation fraction: 30%	Batch size: 32
Early stopping= yes - patience 20	Framework: Torch

Held-out metric gauges

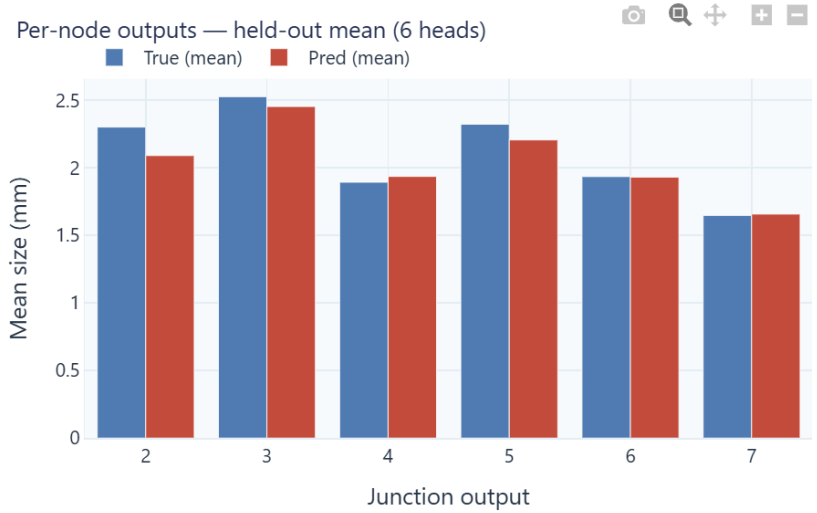


MAE mm

2.212

Per-node output means

6 HEADS



RPTU

Practical Principles to Remember

1. **Physics before learning:** Let hydraulic knowledge guide the machine learning framework.
2. **Features matter:** Invest time in feature engineering and physical interpretation.
3. **Labels define the problem:** Detection, localization, and sizing require clearly defined outputs.
4. **Generalization over memorization:** Good validation performance matters more than low training error.
5. **Data before complexity:** Representative and informative data are more valuable than unnecessarily complex architectures.
6. **Trust but verify:** Check physical consistency, quantify uncertainty, and validate against real measurements.

The objective is not to develop the most complicated neural network. It is to develop the simplest reliable model that extracts the maximum useful information from the physics of the system.

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Thank you!

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